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The Impact of Residential Moves and Union Dissolutions on Children's Educational Attainment

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Summary

This dissertation investigates the impact of family disruptions, residential moves and union dissolutions, on children's educational attainment, using longitudinal data from Danish national registers. The study employs a multiprocess multilevel model with Bayesian estimation to analyse how these disruptions affect the probability of completing upper secondary education. The findings indicate that residential moves have a more substantial negative effect on educational attainment than union dissolutions. However, accounting for endogeneity significantly reduces these effects, particularly for the impact of residential moves. Furthermore, the mother-level random effects from the equation for mothers' residential moves showed the highest correlation with the mother-level random effects from the children's educational attainment, and also as expected had the biggest adjusting effect on the estimates of the impact of disruptions. The correlation between mother-level random effects of residential moves and union dissolutions was also found to be very high, showing how interconnected these processes are, and how they should be accounted for when measuring effects. The research highlights the importance of controlling for unobserved heterogeneity when studying the impact of family disruptions on educational outcomes, offering a more accurate assessment of these relationships.

1 Introduction

Family dynamics have a sinificant impact on children's development, and educational attainment often reflects the broader stability or instability within a household. Among the many factors influencing educational attainment, residential moves and union dissolutions stand out due to their disruptive nature to the child's life.

However, existing literature has predominantly investigated these disruptions in isolation, neglecting their complex correlated effects and the potential selection bias that can distort their estimated impacts. Additionally, the reliance on naive statistical methods does not account for the selection bias that is apparent in these effects.

This thesis examines the effects of residential moves and union dissolutions on children's educational attainment, focusing specifically on secondary school completion. By leveraging longitudinal data from Danish national registers, this study provides a comprehensive analysis of how these family disruptions impact educational outcomes over time. Understanding these effects is crucial, as early educational achievement is a strong predictor of future socioeconomic status and well-being.

The primary aim of this study is to quantify the effects of residential moves and union dissolutions on children's likelihood of completing secondary education. The analysis accounts for the selection bias that is present in these family disruptions by employing a multiprocess multilevel framework with Bayesian estimation. This approach expands the literature as it allows for a more nuanced understanding and disentangling of the complex cross-effects between family disruptions and educational outcomes than previous literature.

The thesis is organised as follows: Chapter 2 reviews the existing literature on family disruptions and educational outcomes, with a particular focus on the methodological challenges associated with measuring these effects. Chapter 3 describes the data used in this study, detailing the construction of the dataset from Danish register data. Chapter 4 outlines the methodological framework, including the multiprocess multilevel modelling approach, and Bayesian estimation techniques. Chapter 5 presents the results for the effects of residential moves and union dissolutions on secondary school completion.

2 Literature Review

In this section, I discuss the prevalence of disruptions of union separations and residential moves for children, and how this can affect their educational outcomes. I also discuss how selection bias affects the estimation of these effects, and how selection bias has been adressed in previous literature. Finally, I review the methodology behind the main methods I am using.

The impact on children's education of the two types of family disruptions, residential moves and union dissolutions, have been well-studied, but typically in isolation. However, an understudied aspect is the correlation between residential moves and union dissolutions, as moves often occur as a consequence of a couple separating (Anderson et al., 2014; Astone & McLanahan, 1994). In Denmark, when a couple separates, it is common for one partner to move out of the shared residence, with the woman usually being the one who relocates. In 80% of cases where children are involved, they move with the mother (Henning Christiansen, 2020). Because these events often happen simultaneously, failing to distinguish between them can lead to confounding, where the effects of one process are incorrectly attributed to the other.

My dissertation adds to the literature in that children's educational attainment is modelled alongside both residential moves and union dissolution processes. This gives a more detailed insight into how these 3 processes are intercorrelated, and what the risks are of not accounting for selection bias.

2.1 The Effects of Disruption on Educational Attainment

Residential Mobility

Residential mobility is not a random process but a highly selective one. Not everyone desires to move, and among those who do, not everyone can. Socioeconomic status (SES) is a significant determinant of residential mobility. Families with lower SES tend to move more frequently, often over shorter distances. Conversely, families with higher SES might move less often, but when they do, it is usually over longer distances (Kull et al., 2016; Nivalainen, 2004; Schacter, 2004).

Furthermore, moving patterns may reflect underlying parental values, such as the preference for stability, which in turn can influence children's educational aspirations and outcomes. Research indicates that family values significantly impact children's school performance,

suggesting that the motivations behind residential moves can have broader implications for children's education (Gorard et al., 2012).

The effect of residential mobility on children is also well-studied. One body of literature identifies a negative relationship between residential moves and children's wellbeing and educational outcomes (Hagan et al., 1996; Ingersoll et al., 1989; Rumberger & Lim, 2008). Scanlon and Devine (2001) highlight several mechanisms that can explain this relationship: Firstly, moving induces stress and strain on the family. The logistics of moving are time-consuming and expensive, and adapting to a new environment and forging new social connections is stressful. Secondly, moving can result in a loss of social capital for both parents and children. When a family relocates, they may lose existing social ties and must establish a new social network in their new community, which can also include change of schools for the children. The disruption to parents' social networks can indirectly affect their children, as parents' social capital often provides resources and opportunities for their children. However, some studies challenges these conclusions, indicating that the impacts are relatively minor when controlling for pre-existing differences between movers and non-movers (Alexander et al., 1996; Garboden et al., 2017; Pribesh & Downey, 1999; Tucker et al., 1998).

On the other hand, some studies find potential positive effects of residential moves (Hango, 2006). Moving can be a strategic decision by parents to improve their circumstances, a phenomenon known as "moving for opportunity." This can indirectly benefit children by enhancing their parents' financial and social situations. It can also have direct benefits, as children may gain access to safer neighborhoods and better schools (Hango, 2006). However, in Denmark, the strategic aspect of moving for better schools or neighborhoods may be less pronounced due to the country's relatively low economic inequality and the more uniform quality of state schools compared to other countries.

Union Dissolutions

Similarly to residential moves, union dissolutions are also highly selective processes that reflect underlying family values and dynamics. These values can also influence the children's educational aspirations and outcome, beyond the direct impact through the instability introduced by the dissolution itself.

The existing literature has predominantly focused on divorce, partly because formal marital relationships are easier to track in data than informal partnerships and cohabitation.

Studies examining the impact of divorce consistently find a "divorce penalty" on children's educational attainment, where children from divorced families tend to achieve lower educational outcomes compared to those from intact families (Bernardi & Radl, 2014; Brand et al., 2019; Gruber, 2004; Rumberger & Lim, 2008). This effect is often attributed to the loss of time and interaction with a parent, typically the father, rather than solely to the disruption caused by the dissolution itself.

Moreover, research has highlighted heterogeneity in the effects of divorce based on socioeconomic status (SES) and parental education. Families with greater resources, such as higher parental education, tend to buffer the negative impacts of divorce more effectively (Bernardi & Radl, 2014; Brand et al., 2019; Guetto et al., 2022; Nilsen et al., 2020). This is often explained by the ability of more resourceful families to provide additional support and maintain stability during and after the divorce process.

However, much of this literature centers on formal marriages, leaving a gap in understanding the effects of separations from non-marital partnerships. The dissolution of informal unions can be just as disruptive to children, as they can involve similar levels of instability and adjustment. Each time a parent enters a new relationship, children must adapt to the presence of a new adult in the household.

The literature on parental remarriage shows varied patterns of impact. Some studies report negative consequences of remarriage on children's wellbeing and educational outcomes, highlighting the challenges children face when integrating a stepparent into their lives (Biblarz & Raftery, 1999; Case et al., 2001; Cassidy et al., 2014). These studies often focus on the reduced time children spend with their biological parent following divorce, typically the father, as a key factor influencing these outcomes. The addition of a stepparent can further complicate family dynamics, leading to additional stress and adjustment challenges for the child. Some literature has found a positive impact of remarriage, explained with the fact that the children gain access to additional resources and capital from the new partner.

2.2 Identification Risks: Mitigating Selection Bias

A main risk to identification of causal effects is endogeneity. Is the effect attributed to the disruption, or is it caused by unobserved factors that affect both the propensity for family disruptions, and the educational attainment? Factors that make a family more prone to disruptions can also influence children's academic achievement.

As reviewed, a big part of previous literature uses simple multiple regression and find a

significant, negative association between residential moves and union dissolutions (most often divorces) and educational outcome for the children. Studies have included a large set of explanatory variables to control for socioeconomic status, children's individual characteristics, etc. to counter selection bias. But as previously discussed, the naive method is vulnerable to endogeneity. Children experiencing disruptions are fundamentally different from children that do not, in both observed and unobserved ways. Even if using rich data with many covariates, the risk of omitted variable bias is present.

One method to counter this is propensity score matching, where children that are similar on observable parameters except for the exposure to disruption. The estimated propensities to experience the disruption is used to reweigh the sample, simulating a counterfactual experiment of what would have happened if these children were not exposed to the event. Studies using this have found a smaller estimate of the effect compared to simple multiple regression. However, propensity score matching remains vulnerable to omitted variables bias, as the students are matched on the observed charactersitics, but not the unobserved, leaving remaining bias in the estimates.

A way to estimate casual effects is to look for 'natural experiments' in society, which create exogenous variance to estimate the causal effect. But finding such sources of exogenous shocks is hard. However, it has been previously done, for example when Gruber (2004) analysed the effect of divorce on different life outcomes for children by exploiting the 'natural experiment' in changes in divorce regulation over time and between states to do a fixed effects study.

Another way to exploit exogenous variance is to use instrumental variables (IV) analysis. But this method requires a valid instrument, which is correlated with the treatment variable (exposure to disruption), and uncorrelated with education except through its correlation with the treatment. Such instruments are hard to find. Cordes et al. (2019) examined residential and school mobility by employing the instrument: Whether families are renting a property that is foreclosed or sold. They still found significant effects, but they found evidence of selection bias as the estimates were smaller than for the naive OLS method.

But as instruments are hard to find, is there another approach that still yields consistent estimates? In the following, I will review the foundational methodology behind my approach in this thesis, and finally discuss multilevel multiprocess modelling.

2.3 Methodological Foundation

This thesis uses a multilevel multiprocess framework with logistic regressison and discrete-time event history analysis. In this section, I briefly review the literature behind these methodologies.

2.3.1 Discrete-time Event History Modelling

Event history modelling is a modelling strategy for analysing the length of time until the occurrence of an event. The dependent variable is the duration until the event happens.

In social research, event history data is often recorded in discretely-measured durations, like annual intervals. Allison (1982) described how basic logistic regression models can be used, treating multiple time units for each individual as individual observations. This also allows for time-varying covariates, and the resulting ML estimators are analogous to the ML estimators of continuous-time data. The following review is also based on Steele and Washbrook (2013) and Steele (2008).

Consider individual j. The observation continues until time t, where either an event occurs, or the observation is right-censored. Right-censoring happens when the event happens outside of the observation period, such that the end time is not known. The variable δ_j is set to 1 if j is uncensored, otherwise it is zero. An observation is censored if the end time is not known, as it is outside of the observation period, or because the individual for some reason is not observed in the entire observation period (for example because of death before the event happened). If assuming that the censoring is non-informative, such that the event times are independent of the censoring mechanism, we can leave the dependent variable, the event indicator, to be 0 at the end of the observation period. Then the binary response variable y_{tj} is defined as

$$y_{tj} = \begin{cases} 0 & t < y_j \\ 0 & t = y_j, \delta_j = 0 \\ 1 & t = y_j, \delta_j = 1. \end{cases}$$

The hazard function for each interval t is defined as the conditional probability of an event during interval t, given that no event has occurred in a previous interval:

$$h_{tj} = Pr\left(y_{tj} = 1 | y_{sj} = 0, s < t\right)$$

The hazard is the usual response probability for a binary variable. The response variable can be analysed using a model suitable for binary responses, for example the logit model:

$$\text{logit}(h_{tj}) = \alpha(t) + \boldsymbol{\beta}^T \boldsymbol{x}_{tj}$$

The baseline log-hazard $\alpha(t)$ is specified as some function of t. Common choices are a stepwise function by treating t as a categorical variable, or a quadratic function defining $\alpha(t) = t + t^2$. $\boldsymbol{x}_{tj}$ is a vector of covariates, which can be fixed or time-varying individual characteristics.

Multilevel Modelling for Event History Data

Commonly in social research, event history data does not have just one episode per individual, but rather a series of them. An example is residential moves, as most people move several times over a lifetime. The repeated events can be viewed as having a two-level hierarchical structure with episodes of exposure to the risk of an event nested within individuals. Because of this hierarchical structure, treating the observations as independent violates the independence assumption of standard regression methods, and it leads to misleading standard errors, as observations within an individual are correlated. These types of hierarchical data can instead be analysed with multilevel modelling by including random effects to capture the underlying individual-level effects.

Consider now individual j experiencing different episodes i, where the individual is at risk of an event. When an event occurs, the duration clock is reset to zero, and a new episode can be started.

$$h_{tji} = Pr\left(y_{tji} = 1, I = i | y_{sji} = 0, s < t\right)$$

$$\text{logit}(h_{tji}) = \alpha(t) + \boldsymbol{\beta}^T \boldsymbol{x}_{tji} + u_j$$

The random effects component captures the unobserved individual-level variance, and is usually assumed to follow a normal distribution $u_j \sim N(0, \sigma_u^2)$. The model assumes that the effects of duration and covariates are constant across individuals, and that the durations of episodes for the same individual are independent, after controlling for u_j.

2.3.2 Multiprocess Modelling

In multilevel modelling, often the assumption that $\boldsymbol{x}_{ij}$ is exogenous, i.e. uncorrelated with both u_j and e_j, is too strong. Unmeasured variables that affect y_{ij} are often also affecting covariates, which makes them endogenous.

Multiprocess modelling handles selection by modelling the outcome equation simultaneously with the endogenous model. In this case, I will discuss a simultaneous equations models framework, where the endogenous variable can be modelled with event history analysis. This yields a system of equations where it is assumed that the unit specific error terms follow a multivariate normal distribution. The correlation between them describes the direction of the bias.

Brien and Lillard (1994) showed how the identification of a multiprocess model using event history models depends on the existence of multiple episodes. Brien and Lillard (1994) used this method when they estimated a joint model of education, marriage, and first conception. They found that while controlling for the fact that couples that cohabitates before marriage are a selective group, the effect on marital dissolution dissipated. Other examples of this methodology applied in literature is Lillard et al. (1995) and Upchurch et al. (2002). The most relevant to my approach is Steele et al. (2013), whom used a multiprocess multilevel model setup to estimate the causal effect of marital dissolutions on children's educational transitions. They found that while correcting for selection bias, the effect of marital dissolution becomes substantially smaller. I use an approach similar to this, but instead of modelling educational transitions, I only focus on one educational outcome: Upper secondary education completion. I also include residential moves in my model to dissect impacts of these two types of disruptions.

2.3.3 Bayesian Estimation, MCMC, and Gibbs Sampling

In Bayesian statistics, the aim is to find the posterior distribution. But doing so is complex, as it involves solving a many parameter integral. However, it can be exploited that it is often easy to simulate values from the conditional posterior distributions. Sampling from the conditional posterior distributions is equivalent to sampling from the joint posterior distributions. This is typically done with Markov Chain Monte Carlo (MCMC) methods. A Markov Chain as every value generated only depends on its previous values through the last value generated. This section is based on Browne (2023).

MCMC methods are increasingly used as a means of modelling complex data structures,

as they are flexible, and increased computational memory and power has made it feasible to use them. The Gibbs sampler (Geman & Geman, 1984) is one of the most widely used algorithms to generate MCMC samples. It is a special case of the general framework of the Metropolis-Hastings algorithm (Hastings, 1970; Metropolis et al., 1953).

Gibbs sampling works by iteratively sampling from conditional distributions of each variable given all the other. For example, consider the simple random effects model for a continuous y:

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + u_j$$

This model has 4 unknown parmaters to estimate: $\beta_0, \beta_1, \sigma_u^2, \sigma_e^2$. The Gibbs sampler then repeatedly samples from the following posterior distributions:

1. $p\left(\beta_0 | y, \beta_1(0), \sigma_u^2(0), \sigma_e^2(0)\right)$ to generate $\beta_0(1)$
2. $p\left(\beta_1 | y, \beta_0(1), \sigma_u^2(0), \sigma_e^2(0)\right)$ to generate $\beta_1(1)$
3. $p\left(\sigma_e^2 | y, \beta_0(1), \beta_1(1), \sigma_u^2(0)\right)$ to generate $\sigma_e^2(1)$
4. $p\left(\sigma_u^2 | y, \beta_0(1), \beta_1(1), \sigma_e^2(1)\right)$ to generate $\sigma_u^2(1)$

When all steps have been performed, all the unknown quantities have been updated. The sampling is then repeated a decided amount of times, using the previously generated set of parameter values to generate the next set. The chain of values generated is the markov chain, where every value generated only depends on its previous values thorugh the last value generated.

This can be adapted for a binary outcome variable y_i by using latent variable formulations (Browne, 2023; Plummer, 2020). It was first proposed by Albert and Chib (1993) for a probit model, and extended by Held and Holmes (2006) to encompass logit models. A latent variable is a continuous variable that is linked to the binary outcome through thresholding. In simple terms, the Gibbs sample algorithm would then sample the latent variable from its conditional distribution, typically a truncated normal distribution if using a probit model. Then y_i is updated based on whether the latent variable exceeds the threshold.

Usually there is no information apriori about the distribution, thus it is most feasible to initialise the model with starting values that reflect this lack of knowledge, i.e. 'flat' or weakly informative priors. In practice, the choice of starting values is not important as the MCMC sampler should be run long enough to 'forget' its initial states. This can be checked

by performing a number of runs with dispersed starting values to check sensitivity to the choice of starting values.

The output of the MCMC sampler should be monitored to check that the chains are mixing well and have converged. This can be done graphically by looking at the trace plots of each parameter (Gilks et al., 1995). It should look like white noise with a stable mean. Furthermore, the Gelman-Rubin convergence diagnostic (Gelman & Rubin, 1992) can be used to assess when the simulation can be stopped. It is generally accepted that the simulation can be stopped when the diagnostic has a value of less than 1.1.

The resulting chain of values from the MCMC sampler are assumed to be from the desired joint posterior distribution. It can then be interpreted by summarising the sequence of posterior estimates. The Bayesian framework makes no assumptions about the distribution of the parameters, and thus it is not suitable to compute p-values or confidence intervals as per the frequentist approach. Instead, the means, standard deviation, and credible intervals (CI) should be interpreted. The credible interval is a quantile based estimate and is simply the desired quantiles taken from the sequence of generated values, not calculated based on a ditribution assumption as in the frequentist approach.

3 Data

3.1 Danish Register Data

This dissertation utilises administrative register data from Statistics Denmark, encompassing different registers on various sociological characteristics, address changes, and family types. The study includes children born in Denmark between 1986 and 2001, and their mothers. The starting year, 1986, was constrained by the detailed data on mothers' unions, which is available from 1986. The endpoint, 2001, is constrained to allow for the tracking of educational outcomes up to age 20, as the newest data on completed education is available for year 2021. In this time, almost all of the children are registered with a mother and a father.

I included all the mothers' children born in the time period of 1986-2001, excluding any born outside of this interval. This means that some sibships are reduced in size, as siblings born before 1986 or after 2001 are not included to ensure a similar data basis for all children included. The average children pr. woman is 1.72 in this sample, but the real number of average children pr. woman, including those outside of this time interval is 2.40.

In total, this dataset encompasses 599,219 mothers and 1,034,573 children. Given the size of the dataset, the computational complexity of fitting hierarchical Bayesian models, and the project's limited scope, a random subsample of 11,532 mothers and their 19,790 children from the time period 1986-2001 was selected due to computational constraints to reduce model run time. The mothers had 59,620 residential moves, and 21,680 unions.

The presented summary statistics and results are based on this sample. This sample was checked for and showed no significant deviation from the overall population characteristics. Initially, 12,000 mothers were chosen, but 4.2% of the mothers were removed due to all her children having an unknown father or missing educational data, in which case they were omitted from the sample.

3.2 Secondary education in Denmark

Completion of secondary education is a commonly used indicator variable in social research, as early educational achievement is a strong predictor of future socioeconomic status and well-being (Garboden et al., 2017; Jæger & Holm, 2007). Upon completing the mandatory nine years of elementary school at around age 15, Danish children choose between three paths: leaving the educational system, entering upper secondary education, or pursuing

vocational secondary education. The choice of secondary education is essential as it significantly influences future educational opportunities due to the structured nature of the Danish education system. Upper secondary education, known as "gymnasium", is the academic track and typically takes three years to complete. This path grants access to higher education, such as university degrees, which about 70% of high school students pursue. The amount of students on upper secondary education is increasing, and in 2018, approximately 70% of each cohort opts for upper secondary education. (Jæger & Holm, 2007; StatisticsDenmark, 2018).

3.3 Family disruptions

Family disruptions are tracked by annual changes, with status checks conducted every January 1st to identify changes from the previous year.

This approach means that if multiple disruptions occur within a single year, only the latest one is recorded. For example, if a mother moves twice in one year, only the last move is captured. Similarly, if a mother separates from a partner in the beginning of a year but reconciles before the year ends, she is recorded as having maintained the union. This leads to underestimation of the family disruptions, especially of the mothers that frequently move and dissolve unions.

The data is right censored under two circumstances: The last observed union remains intact or if the last address remains unchanged at the end of the observation period, or if the last move or union dissolution is caused by the mother's death. There are also left censored observations in the case that the union or move is initiated before 1986. In that case, the union or move is omitted.

Residential moves

Residential moves are documented using the Danish central address registers (`BEFBOP`, `BEFADR`), which captures all address changes in Denmark by the exact date of address change. For simplicity and to keep consistency with the union data set, I consider only the address at January 1st every year.

Union dissolution

Union dissolutions are recorded using data from the Danish population register (`BEF`), which registers all people with a Danish address January 1st since 1986. I follow Statistics Denmark's definition of family types (`familie_type`) StatisticsDenmark (2024a) to define a

union. According to these guidelines, two adults are considered a couple if they reside at the same address and meet any of the following criteria:

1. They are married or in a registered partnership
2. They have at least one common child
3. They are not close family, are of different genders, and have an age difference of less than 15 years

This is a broader definition of unions than what is commonly used in previous literature, which often focuses solely on formal marriages. This broader definition is particularly relevant in a country like Denmark, where a significant proportion of partnerships are not officially registered. Only 59 % of children in Denmark live in a household where the adults are married, while the remainder live with either a single parent or in a cohabiting relationship with unmarried partners (StatisticsDenmark, 2024b).

It is crucial to include informal unions in the analysis, as the presence of a new partner, regardless of marital status, introduces another adult who spends a considerable amount of time with the child, influencing both the household dynamics and the child's development. Neglecting to account for these informal unions would lead to an underestimation of the family disruptions that children experience.

It is not uncommon for adults to share housing at the same address, which could lead to some roommate pairs being mistakenly classified as partnership unions. However, this is most common for young adults who do not yet have children.

3.4 Multilevel structure of data

The data is organised in a multilevel structure, with mother identification numbers serving as the level 2 units. For the education analysis, the level 1 observations are her children. For the event history models, the level 1 observations are each mother's repeated annual records. This relationship is visualised in fig. 1.

On average, each mother has 1.72 children, lives at 5.17 addresses, and has 1.88 unions. Detailed distributions of these metrics are presented in fig. 2. Gaps in observation for residential moves may arise if a mother is not registered with an address in Denmark, typically due to moving abroad, which is not common. In contrast, gaps in observations for unions is more common for mothers to live alone in a single parent household at least for periods of time following a divorce or separation. The residential moves dataset has on

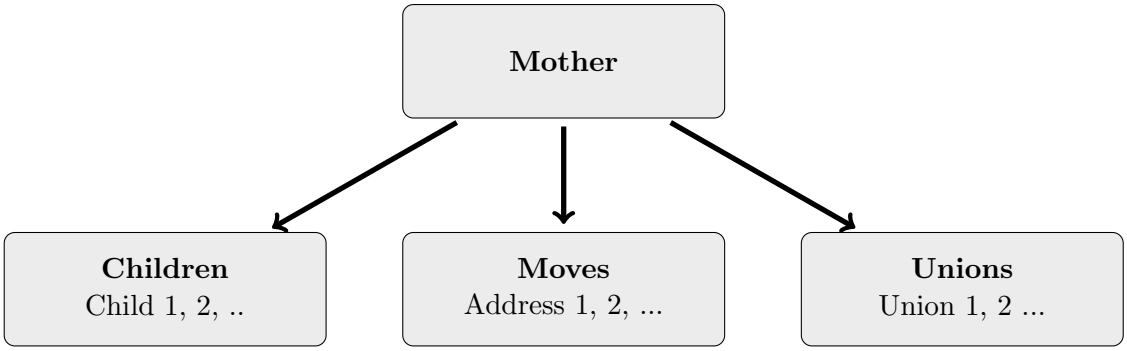


Figure 1 Diagram of the multilevel structure of the data.

average 31 observations per mother, and the average number of observations per mother for union dissolutions is 26.

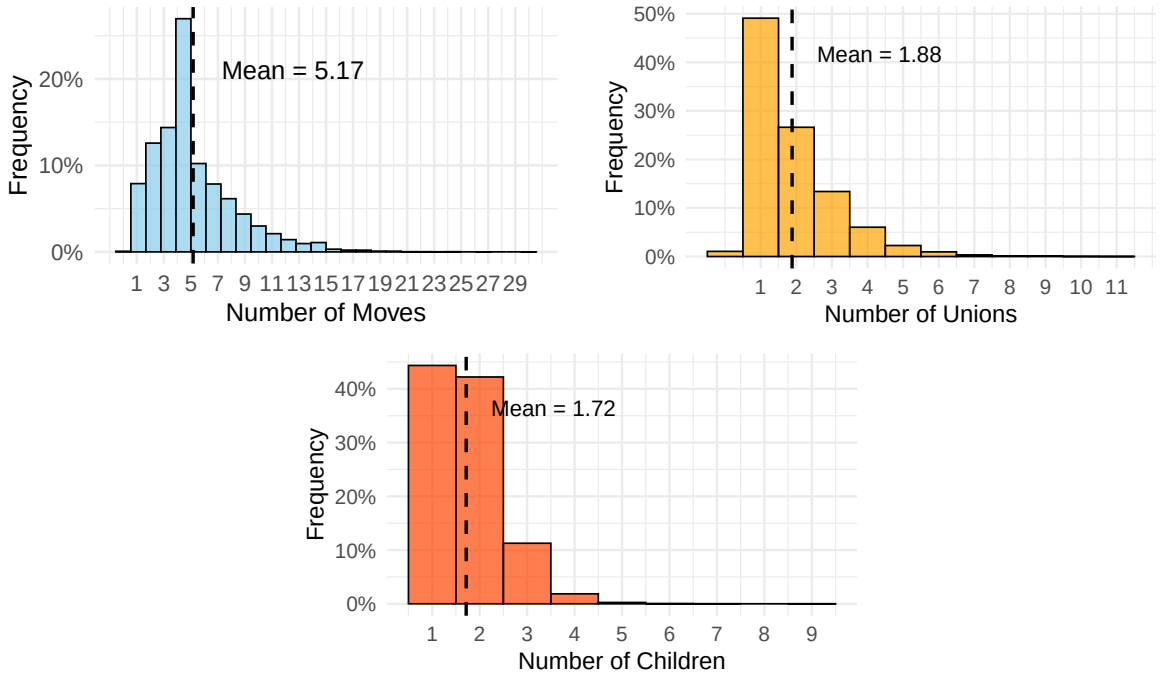


Figure 2 Histograms showing the distribution of number of moves, unions, and children per mother.

3.5 Variables

table 2 shows descriptive statistics for the variables used in the analysis of the three equations. Each set of variables is described below in greater detail. Variables were chosen based on the literature review to find commonly used variables that I could access through Statistcs Denmark. See section 5.1 for a more detailed discussion on the inclusion and exclusion of variables in the model.

Variables for the educational attainment model

Outcome: *Completed upper secondary education.* Binary indicator if the child has completed upper secondary education at age 20.

Variable of interest: *Residential moves.* The total number of residential moves made by the mother up until the child's 18th year of life.

Variable of interest: *Union dissolutions.* The total number of the mother's union dissolutions up until the child's 18th year of life.

Boy. Boys are coded 1 and girls are coded 0.

Birthyear. I include a birthyear variable to account for the effect of younger cohorts completing upper secondary education more often.

Parents never married. Binary indicator if the child's biological parents never married.

Mother's age at birth. To not assume a linear effect, I created categorical variables for the mother's age when the child was born. This was based on preliminary analysis of the data, looking at patterns of mother's age at birth and the child's upper secondary school completion. Categories chosen are up to 21 years, 22-27 years, 28-33 years, and 34+ years old.

Mother's education. I considered the years of completed education when the child turned 20. I then translated this into 4 categories based on the DISCED standard: Primary (Up to 13 years of schooling), secondary (13-15 years), Diploma/Bachelor (15-18 years), and Postgraduate (18+ years). Although the data quality is high, a small percentage of cases have missing values. Preliminary analysis showed that the missing data had similar coefficients to the primary school category, so they were included in that group.

Father's education. Similarly for the mother's education, just for the child's registered father.

Variables for the event history models

Outcome: *Residential move / Union dissolution* Binary indicator if the mother had an event in the given year.

Year. The year of the observation

Move count/Union count. the cumulative count of unions or moves. The first episode has the value 1, the second 2, and so on.

Duration. The cumulative duration in years of the current episode.

Variable	**Mean**	**SD**	**Min – Max**
Completed upper secondary education	57.60%		
Residential moves			
1	24.97%		
2	18.15%		
3	11.93%		
4	7.91%		
5+	12.37%		
Union dissolutions			
1	27.61%		
2	10.13%		
3	3.35%		
4+	1.42%		
Boy	51.25%		
Birthyear	1992.23	4.51	1986 – 2001
Parents never married	9.10%		
Mother's birth age			
22-27 years	39.18%		
28-33 years	39.67%		
≥34 years	13.01%		
Mother's education			
Diploma/Bachelor	26.56%		
Secondary	40.16%		
Postgraduate	6.82%		
Father's education			
Diploma/Bachelor	20.13%		
Secondary	47.02%		
Postgraduate	9.73%		

Table 1: Descriptive statistics for variables included in the model for children's education

Age at move-in/start of union. The mother's age at the start of the episode.

Mother's education. The mother's education level in the given year, defined as in the children's education outcome model.

Variables only included for residential moves

Copenhagen. Binary indicator if the current address is Copenhagen.

Variables only included for union dissolutions

Partner's education. The partner's education level in the given year, defined as in the children's education outcome model.

Married. Binary indicator if the mother is married to her current partner.

Number of children with partner. The number of children that the two people are both

Variable	Mean	SD	Min	Max
Completed upper secondary education	57.60%			
edu$_m ove_g roup_1$	24.97%			
edu$_m ove_g roup_2$	18.15%			
edu$_m ove_g roup_3$	11.93%			
edu$_m ove_g roup_4$	7.91%			
edu$_m ove_g roup_5 p$	12.37%			
edu$_u nionbreak_g roup_1$	27.61%			
edu$_u nionbreak_g roup_2$	10.13%			
edu$_u nionbreak_g roup_3$	3.35%			
edu$_u nionbreak_g roup_4 p$	1.42%			
male	51.25%			
birthyear	1992.23	4.51	1986	2001
parents$_n evermarried$	6.1%			
edu$_m other_b irthagegroup_2 2_2 7$	39.18%			
edu$_m other_b irthagegroup_2 8_3 3$	39.67%			
edu$_m other_b irthagegroup_g eq 34$	13.01%			
edu$_m other_e ducat_D iploma_B achelor$	26.56%			
edu$_m other_e ducat_P ostgraduate$	6.82%			
edu$_m other_e ducat_S econdary$	40.16%			
edu$_f ather_e ducat_D iploma_B achelor$	20.13%			
edu$_f ather_e ducat_P ostgraduate$	9.73%			
edu$_f ather_e ducat_S econdary$	47.02%			

Table 2: Descriptive statistics for variables included in the model for children's education

parents to.

Children with previous partner (Mother). Binary indicator if the mother has children with another person when she entered the union.

Children with previous partner (Partner). Binary indicator if the partner has children with another person when they entered the union.

4 Methodology

This study applies a multilevel multiprocess framework similar to i.e. Lillard et al. (1995) and Steele et al. (2009) to analyze the complex effects of family disruptions on children's educational outcomes, specifically residential moves and union dissolutions. I jointly estimate 3 equations to allow for correlation between the mother-level random effects: A logit model for the children's educational outcome, and discrete-time event history models for the mother's residential moves and union dissolutions. This setup is made possible by the fact that mothers might have more than one child, live at different addresses, and form different unions over a lifetime. The setup makes it possible to capture the underlying, unobserved factors that influences all three aspects simultaneously. An estimated non-zero correlation between the random effects affecting the child's educational outcome model, and the random effects affecting the mother's risk of partner separation or moving signals endogeneity in the relationship between the child's educational outcome and experienced family disruptions.

Understanding these effects can be challenging because there are often unmeasured characteristics that simultaneously influence the likelihood of family disruptions and educational achievements. This predisposes some children to both family instability and educational challenges and confounds the observed relationship. These hidden factors might include familial attitudes that deprioritize stability, thus increasing the propensity for parental separation and frequent relocations, while also undervaluing education. Alternatively, inherited traits such as psychological resilience could predispose families to instability and affect children's ability to succeed academically. Figure 3 visualises this relationship.

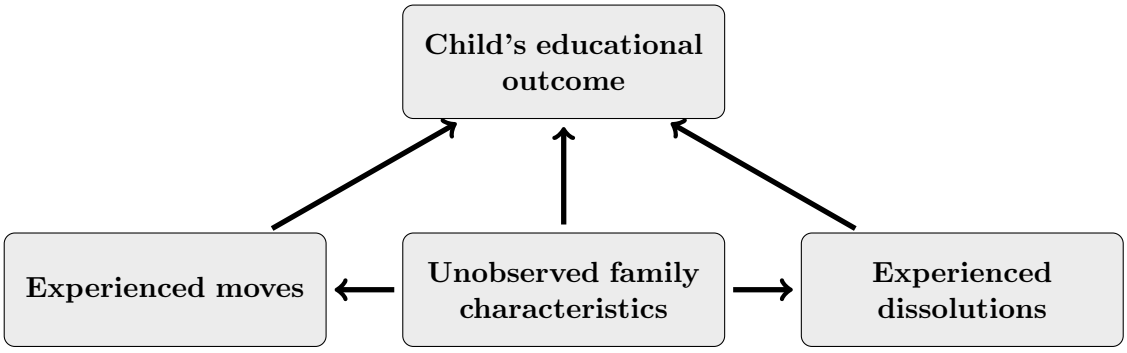


Figure 3 Diagram of the relationship between family characteristics, experienced moves, experienced dissolutions, and child's educational attainment.

I assume the mother to be the primary conduit for unmeasured family characteristics

influencing the child, which aligns with research findings that a mother's traits often have a greater impact on a child's educational outcomes compared to the father's. This includes factors such as parental education levels and health issues. Additionally, following a separation or divorce, children are more likely to reside with their mother rather than their father (Henning Christiansen, 2020), thereby exposing them more directly to any changes in their mother's life, such as relocation and new unions.

In section 5.6, I discuss potential remaining sources of bias that can lead to identification issues.

4.1 Children's Educational Outcome

Let y_{ij} be a binary outcome variable coded 1 if child i of mother j finishes high school. $p_{ij}^E = P(y_{ij} = 1)$. To model this probability, the logit link function is applied, ensuring that the probability lies between 0 and 1:

$$\text{logit}(p_{ij}) = \log\left(\frac{p_{ij}}{1 - p_{ij}}\right)$$

The specific model specification for the effect of family disruptions on the child's educational outcome is:

$$\text{logit}(p_{ij}^E) = \boldsymbol{\beta}^E \mathbf{X}_{ij}^E + \boldsymbol{\gamma}\text{moves}_{ij} + \boldsymbol{\lambda}\text{unions}_{ij} + u_j^E \tag{1}$$

Here, $\boldsymbol{\beta}^E$ is a vector of coefficients for the covariates $\mathbf{X}_{ij}$, which include characteristics of both the child and parents. The terms $moves_{ij}$ and $unions_{ij}$ are vectors indicating the degree of disruptions experienced by the child with vector of coefficients γ and λ, respectively. The random effect $u_j^E \sim N(0, \sigma^2)$ captures unobserved heterogeneity at the mother level.

4.2 Mothers' Risk of Residential Moves and Union Dissolutions

To model the mother's risk of residential moves andcunion dissolutions, I use a discrete-time multilevel event history model for repeated events similarly to Lillard et al. (1995), and Steele et al. (2009).

Define an *episode* as a period for which an individual is at risk of experiencing an *event*, i.e.:

- **Residential move**: An episode is the duration an individual resides at a particular address, with the event marked by a change in address.

- **Union dissolution**: An episode is the duration of cohabitation with a partner, with the event being that they move apart.

When an event occurs, the duration clock resets to zero, marking the beginning of a new episode when the mother relocates or forms a new union.

A feature of the data is the repeatable nature of these events, where mothers typically move more than once and have more than one union. Thus, the data exhibits a two-level hierarchical structure, with episodes (level 1) nested within mothers (level 2), and the duration of events for the same woman might be correlated. This justifies the use of random effects in modeling.

I define a binary response $y_t ji$ to indicate if an event happens in a certain year for each episode j for individual i, with the corresponding hazard functions h_{tij}^M and h_{tij}^U. The probability of the event happening, the discrete-time hazard, is denoted π_{ijt}^M and π_{ijt}^M. The logit link function is used again. The individual equations are given below:

Residential moves

$$\text{logit}(\pi_{tij}^M) = \alpha^M D_{tij}^M + \boldsymbol{\beta}^M \mathbf{X}_{ij}^M + u_j^M \tag{2}$$

Union dissolutions

$$\text{logit}(\pi_{tij}^U) = \alpha^U D_{tij}^U + \boldsymbol{\beta}^U \mathbf{X}_{ij}^U + u_j^U \tag{3}$$

The baseline hazards $\alpha^M D_{tij}^M$ and $\alpha^M D_{tij}^U$ are defined as quadratic functions of time:

$$\alpha D_{tij}^M = \alpha_0^M + \alpha_1^M t + \alpha_2^M t^2$$
$$\alpha D_{tij}^U = \alpha_0^U + \alpha_1^U t + \alpha_2^U t^2$$

$y_{tij}^{(p)}$ is a binary outcome variable, indicating if a union dissolves in year t, $\mathbf{D}_{tij}^{(p)}$ is the cumulative duration of the partnership in years, $n_{tij}^{(p)}$ is a cumulative counter indicating what number the current partnership is for the mother, i.e. 1 for the first partnership and 2 for the second. $\mathbf{x}_{tij}^{(p)}$ is a vector of covariates, including age, education, and the number of a couples' common children. $v_j^{(p)} \sim N(0, \sigma^2)$ are the random effects on the mother level.

4.3 Correlation Structure

The models are linked by allowing each pair of the mother-level random effects to be correlated. Initially, I assume that the correlations between the random effects are zero, i.e. $\sigma_{u(E,M)} = \sigma_{u(E,U)} = \sigma_{u(M,U)} = 0$, which is equivalent to fitting each of eq. (1), eq. (2), and eq. (3) separately. I then relax this assumption by allowing the correlations to be non-zero.

In the full model, the random effects are allowed to correlate freely, assuming the mother-level random effects follow a trivariate normal distribution $\mathbf{u}_j = [u_j^E, u_j^M, u_j^U]^T \sim N(\mathbf{0}, \boldsymbol{\Omega_u})$ with covariance matrix $\boldsymbol{\Omega}_u$:

$$\boldsymbol{\Omega}_u = \begin{pmatrix} \sigma^2_{u(E)} & & \\ \sigma_{u(E,M)} & \sigma^2_{u(M)} & \\ \sigma_{u(E,U)} & \sigma_{u(M,U)} & \sigma^2_{u(U)} \end{pmatrix} \tag{4}$$

The elements of this covariance matrix $\boldsymbol{\Omega}_u$ are used to calculate the correlations between the random effects. For instance, the correlation between the random effects for the child's educational outcome and the mother's residential moves can be calculated as:

$$\rho_{u(E,M)} = \frac{\sigma_{u(E,M)}}{\sqrt{\sigma^2_{u(E)} \cdot \sigma^2_{u(M)}}}.$$

4.4 Bayesian Estimation

The complexity of the hierarchical model prevents it from being fitted using standard software for random effects models. Instead, Bayesian estimation methods are used due to their flexibility in handling complex models.

Bayesian estimation is performed using Markov Chain Monte Carlo (MCMC) methods, specifically Gibbs sampling, as implemented in JAGS (Just Another Gibbs Sampler) Plummer (2020), an engine for running BUGS models. The `runjags` package Denwood (2023) is used to facilitate the process, offering enhancements to the standard `JAGS` implementation. An annotated code example is found in the appendix.

The following non-informative priors were chosen to reflect minimal prior knowledge about the parameters, following a similar approach to Browne (2023) and Steele and Grundy (2021):

- **Regression coefficients:** The priors for the regression coefficients associated with the educational, residential move, and union dissolution models are specified as normal

distributions with mean 0 and variance 0.001, representing a non-informative prior: $N(0, 0.001)$

- **Random Effect Precisions (No Correlation Assumed):** When no correlation between random effects is assumed, the random effect precisions are assigned Gamma priors: $\sigma^{-2}_{u(E)}, \sigma^{-2}_{u(M)}, \sigma^{-2}_{u(U)} \sim \text{Gamma}(0.001, 0.001)$
- **Random Effects Precision Matrix (Correlation Assumed):** For models where correlation between random effects is allowed, the random effects precision matrix follows a Wishart prior: $\boldsymbol{\Omega}_u^{-1} \sim \text{Wishart}(\nu, \boldsymbol{S}^{-1})$ with degrees of freedom $\nu = R$ and scale matrix $\boldsymbol{S} = \boldsymbol{I}_R$. The value of R is 2 when allowing correlation between 2 random effects and 3 when allowing correlation between 3 random effects.

4.5 Data Formatting

Following the approach of chapter 6 from Gilks et al. (1995), I center the numerical variables around their means to improve mixing.

As described in Lunn et al. (2012), hierarchical models pose a challenge in the BUGS language as the batch of data from each equation contains different variables. I solve this by stacking the data from all 3 models (children's education, mother's residential moves, and mother's unions). The binary responses $(y^{(E)}_{ji}, y^{(M)}_{tji}, y^{(U)}_{tji})$ are stacked into a single reponse vector `y`. The different covariates are also stacked, creating a ragged array which is padded out with zeros. I create an indicator variable for each response, which is interacted with the set of explanatory variables for each equation. A simplified example of the data structure is shown in table 3 with just one covariate and 2 rows pr. model. This approach ensures correct inference, but alternatively, an approach based on offsets and nested indexing has computational advantages as described by Lunn et al. (2012).

y	Edu_data	Move_data	Union_data	Birthyear	Move_year	Union_year
1	1	0	0	1989	0	0
1	1	0	0	1996	0	0
0	0	1	0	0	1987	0
0	0	1	0	0	1988	0
0	0	0	1	0	0	1990
1	0	0	1	0	0	1991

Table 3: Formatting of data to comply with the BUGS language.

The code snippet below shows how this is implemented in the JAGS file. The full JAGS model is found in appendix.

```
for (i in 1:N) {
  y[i] ~ dbern(p.y[i])
  logit(p.y[i]) <-
    b.edu_data[1]*edu_data[i] +
    b.edu_data[2]*edu_data[i]*edu_move_group_1[i] +
    b.move_data[1]*move_data[i] +
    b.move_data[2]*move_data[i]*move_year[i] +
    b.union_data[1]*union_data[i] +
    b.union_data[2]*union_data[i]*union_year[i]
}
```

Listing 1: Code snippet showing how the hierarchical model is defined in JAGS.

5 Results

This section is divided into parts. First, I discuss the model selection process. I then present and discuss the findings when including different combinations of correlations of random effects.

5.1 Model Selection

During preliminary analysis, a number of different modelling approaches were explored to determine the most appropiate model setup.

Children's Educational Outcome

To avoid assuming linear effects, I use a vector of categories indicating the exposure to family disruptions. The cut-offs of 5+ and 4+ were chosen such that each category included a sufficient number of children (at least 1%). This decision was based on the limited sample of 11,532 mothers and 19,790 children, making more extreme categories result in unstable estimates that were prone to outlier bias. In a larger sample, I expect that the disruption categories could be expanded to further explore the more extreme cases of family disruptions, particularly for residential moves.

As previous literature has found that the effect of family disruptions on educational outcomes differs for ethnicity, gender, age, geography, and socioeconomic status (Bernardi & Radl, 2014; Ingersoll et al., 1989; Kull et al., 2016; Nivalainen, 2004), these channels of heterogeneity were explored. However, I did not find substantial evidence of heterogeneity effects for these variables, so they were left out for a simpler model. However, I did observe potential age-related heterogeneity, with disruptions having different effects depending on whether they occurred during early childhood (ages 0-5), middle childhood (ages 6-13), or adolescence (ages 14-18). Due to the limited sample size, these effects were unstable, so they were not included. Education level of parents and the gender of the child are instead included as main effects. Geography (Danish regions and municipalities) and ethnicity were not significant predictors and were omitted from the model. Additionally, I explored the interaction between disruptions, to assess whether simultaneously happening moves and union dissolutions would have a bigger impact, but found no evidence to support this.

Considering the literature on the differing impacts of short versus long distance moves, particularly because of school mobility (Pribesh & Downey, 1999; Rumberger & Lim, 2008), I explored the division of residential moves between within-municipality and

cross-municipality, as well as the geographical distance between addresses. However, these factors did not provide sufficient evidence to be included in the model.

Event History Models

For the event history models, I chose to use polynomial terms instead of categorical variables to represent episode count, duration, and year effects. While it is common to use categorical variables in order to not assume a specific distribution, the choice of polynomials was made to simplify the model and reduce the parameter space, which improved computational efficiency. This decision aligns with recommendations by Gilks et al. (1995), who discuss how models with high-dimensional parameter spaces can lead to slower and less stable convergence in MCMC methods. Visual inspection of the data indicated that a quadratic fit was appropriate. However, another trade-off is that it is less intuitive to interpret quadratic terms.

In line with findings from previous studies, I considered including ethnicity and geography as predictors in the event history models (Garboden et al., 2017), but they turned out to be weak predictors. The only exception was an indicator variable for mothers living in Copenhagen, which proved to be a strong predictor of residential moves, likely due to the distinct characteristics of the Copenhagen housing market compared to the rest of Denmark.

5.2 Evidence For Selection: Estimation of Random Effects Variance and Correlations

I fit eq. (1), eq. (2), and eq. (3) in 4 different models, each with different combinations of correlations allowed by specifying the covariance matrix eq. (4). The results are based on two parallel chains of 4,000 MCMC iterations with a burn-in sample of 2,000. The chains are initialised with different sets of starting value and random number generators. A simple example of the code is found in the appendix. Convergence checks for these models indicate stability, including looking at the trace plots and calculating the Gelman-Rubin test statistic which was close to 1 for all parameters.

- Model 1: Independent estimation of all equations (i.e., no correlations allowed between random effects)

  $\sigma_{u(E,M)} = 0, \sigma_{u(E,U)} = 0, \sigma_{u(M,U)} = 0$

- Model 2: Correlation allowed between random effects from children's education and mothers' residential moves

  $\sigma_{u(E,M)} \neq 0, \sigma_{u(E,U)} = 0, \sigma_{u(M,U)} = 0$

- Model 3: Correlation allowed between random effects from children's education and mothers' union dissolutions

  $\sigma_{u(E,M)} = 0, \sigma_{u(E,U)} \neq 0, \sigma_{u(M,U)} = 0$

- Model 4: Correlation allowed between all three equations

  $\sigma_{u(E,M)} \neq 0, \sigma_{u(E,U)} \neq 0, \sigma_{u(M,U)} \neq 0$

Table 4 presents the estimated variances and correlations of the random effects for model 2, 3, and 4.

	Model 2 $\sigma_{u(E,M)} \neq 0$, $\sigma_{u(E,U)} = 0, \sigma_{u(M,U)} = 0$		**Model 3** $\sigma_{u(E,M)} = 0$, $\sigma_{u(E,U)} \neq 0, \sigma_{u(M,U)} = 0$		**Model 4** $\sigma_{u(E,M)} \neq 0$, $\sigma_{u(E,U)} \neq 0, \sigma_{u(M,U)} \neq 0$	
	Mean	**95% CI**	**Mean**	**95% CI**	**Mean**	**95% CI**
Random effect std. deviations						
$\sigma_{u(E)}$	**1.43**	(1.345;1.511)	**1.37**	(1.274;1.446)	**1.36**	(1.248;1.457)
$\sigma_{u(M)}$	**0.41**	(0.378;0.434)	-	-	**0.54**	(0.516;0.556)
$\sigma_{u(U)}$	-	-	**1.41**	(1.332;1.483)	**1.52**	(1.457;1.590)
Correlations						
$\rho_{u(E,M)}$	**-0.24**	(-0.305;-0.160)	-	-	**-0.26**	(-0.317;-0.196)
$\rho_{u(E,U)}$	-	-	**-0.08**	(-0.133;-0.029)	**-0.19**	(-0.241;-0.135)
$\rho_{u(M,U)}$	-	-	-	-	**0.92**	(0.897;0.933)

Table 4: Mother-level random effect parameters from models 2, 3, and 4. Parameters where zero is not included in the 95% CI are in bold.

Model 1, where the equations are estimated independently, indicates significant unobserved heterogeneity at the mother level in children's educational attainment, mothers' risk of residential moves, and union dissolutions. This conclusion is based on the substantial estimated variances in the mother-level random effects, all of which exclude zero in their 95% CI's. The estimated variances remain stable for the subsequent models that include correlations between random effects.

For models 2, 3, and 4, unobserved heterogeneity persits, as evidenced by the significant variances in the random effects. Introducing correlations between education and either residential moves (Model 2) or union dissolutions (Model 3) reveals a residual negative correlation between children's educational attainment and the mother's risk of these disruptions. The effect is notably stronger for residential moves ($\rho_{u(E,M)=-0.24}$) than for union dissolutions ($\rho_{u(E,U)=-0.08}$), with the latter just barely excluding zero in its 95% CI. For the full model 4, the correlation between education and residential moves random effects is largely unchanged, but the correlation between education and union dissolutions

increases to -0.19, possibly driven by the high correlation ($\rho_{u(M,U)}$=0.92) between the mother-level random effects for moves and dissolutions.

The negative correlations signals that children whose mothers are more prone to frequent moves or dissolutions tend to have lower chances of completing upper secondary education. This finding aligns with previous research that has identified an association between family instability and lower educational outcomes. This association means that estimates that are not accounting for selection bias will overestimate the negative effect of residential moves and union dissolutions on educational attainment.

The high positive correlation ($\rho_{u(M,U)}$=0.92) between the mother-level random effects for moves and dissolutions shows that the mothers most likely to move are also most likely to dissolve unions, perhaps due to a dispreference for stability. This is also aligned with literature (cite). However, this exceptionally high correlation could be driven by data definitions, as the definition of unions is based on shared address in the registers section 3.3. Thus, a union is formed the year that the couple moves to the same address, and it is dissolved the year that they move apart, making some of these episodes exactly the same for the residential moves and union dissolutions, which can leave similar error terms bound to the individual mothers. However, in Denmark, following a union dissolution, the mother is only slightly more likely to be the one that moves, rather than her partner (Henning Christiansen, 2020), and as a mother on average has 5.17 different addresses compared to 1.88 unions, a majority of resdiential moves throughout the lifetime happens independently of a union dissolution. The high correlation is still odd and could be investigated further, but it has been stable throughout all analyses with different model specifications used in the exploratory analysis, and it is stable for different starting values of the precision matrix, like initialisng with the estimated variances from separate maximum likelihood estimations of the models. The trace plots also showed adequate mixing, as seen in the appendix.

5.3 Effect of Family Disruptions on Children's Educational Attainment

Table 5 and table 6 show the estimated impact of family disruptions on the child's educational attainment for the 4 different models with varying random effects covariance structures.

In model 1, where random effects are assumed uncorrelated, the impact of both residential moves and union dissolutions on secondary school completion is generally increasing with the number of disruptions. Union dissolutions exhibit a stronger negative effect than residential moves.

	Model 1 $\sigma_{u(E,M)} = 0$, $\sigma_{u(E,U)} = 0, \sigma_{u(M,U)} = 0$		Model 4 $\sigma_{u(E,M)} \neq 0$, $\sigma_{u(E,U)} \neq 0, \sigma_{u(M,U)} \neq 0$	
	Mean	**95% CI**	**Mean**	**95% CI**
Residential moves				
0 (Reference)	-	-	-	-
1	0.07	(-0.032; 0.174)	0.10	(-0.021; 0.216)
2	**-0.19**	(-0.320; -0.088)	-0.10	(-0.245; 0.036)
3	**-0.32**	(-0.479; -0.167)	**-0.19**	(-0.341; -0.027)
4	**-0.24**	(-0.463; -0.049)	-0.14	(-0.242; 0.012)
5+	**-0.61**	(-0.818; -0.417)	**-0.25**	(-0.475; -0.032)
Union dissolutions				
0 (Reference)	-	-	-	-
1	**-0.32**	(-0.428; -0.214)	**-0.26**	(-0.367; -0.150)
2	**-0.43**	(-0.597; -0.256)	**-0.27**	(-0.419; -0.110)
3	**-0.58**	(-0.856; -0.322)	**-0.44**	(-0.720; -0.164)
4+	**-1.18**	(-1.531; -0.805)	**-0.66**	(-1.097; -0.250)

Table 5: Estimated effects of family disruption on secondary school completion for models 1 and 4, where no correlation and free correlation allowed between random effects, respectively. Parameters where zero is not included in the 95% CI are in bold.

In model 4, which accounts for correlations between all three sets of random effects, the estimated effects of residential moves diminish. There is no significant impact for 1, 2, or 4 moves compared to zero moves, while the effects for 3 and 5+ moves remain, though the evidence is weak given that zero is nearly included in the 95% CI's. The effect of union dissolutions stay significant in model 4, though the magnitude decreases after accounting for selection bias. While multiple union dissolutions consistently reduce the probability of secondary school completion, the differences in impact between 1, 2, 3, and 4+ dissolutions are not statistically significant, as the 95% CI's overlap, whereas this was not the case for model 1.

These findings are all in the expected direction as the correlations of the random effects in table 4 were negative, suggesting that a part of the effect of moving and union dissolutions on educational attainment is explained by selection. After controlling for the selection bias, the effect size decreases, although for union dissolutions it remains substantive, the effect largely disappears for residential moves except in cases of extreme frequency of 5+ moves.

Table 6 shows the estimated effect for model 2 and 3, where correlation is allowed between the education equation and residential moves or union dissolutions, respectively.

Model 2 includes the correlation of random effects from the mother's residential moves model, which makes the effect of residential moves decrease to a similar magnitude as when all

	Model 2 $\sigma_{u(E,M)} \neq 0$, $\sigma_{u(E,U)} = 0, \sigma_{u(M,U)} = 0$		Model 3 $\sigma_{u(E,M)} = 0$, $\sigma_{u(E,U)} \neq 0, \sigma_{u(M,U)} = 0$	
	Mean	**95% CI**	**Mean**	**95% CI**
Residential moves				
0 (Reference)	-	-	-	-
1	**0.13**	(0.025;0.240)	0.07	(-0.021;0.174)
2	-0.07	(-0.210;0.072)	**-0.16**	(-0.268;-0.035)
3	-0.01	(-0.170;0.169)	**-0.19**	(-0.336;-0.050)
4	-0.16	(-0.135;0.207)	-0.18	(-0.147;0.210)
5+	-0.22	(-0.445;0.024)	**-0.58**	(-0.740;-0.381)
Union dissolutions				
0 (Reference)	-	-	-	-
1	**-0.22**	(-0.323;-0.128)	**-0.20**	(-0.319;-0.097)
2	**-0.43**	(-0.591;-0.281)	**-0.40**	(-0.583;-0.221)
3	**-0.52**	(-0.793;-0.266)	**-0.71**	(-0.972;-0.405)
4+	**-0.67**	(-1.085;-0.256)	**-0.80**	(-1.247;-0.371)

Table 6: Estimated effect of family disruption on secondary school completion for models 2 and 3, where correlation is allowed between the equation for the child's education and mother's moves or unions, respectively. Parameters where zero is not included in the 95% CI are in bold.

correlation parameters are freely estimated in model 4. The effect sizes are a bit lower than in model 4, but there is no evidence that they are different as the 95% CI's overlap by a big margin. The effect on union dissolutions, however, is interesting. There is still evidence that the effect sizes are different from zero, i.e. that 1, 2, 3, and 4+ experienced union dissolutions lowers the child's probability of completing secondary school, compared to a child that experiences none. The estimated effect sizes are a bit bigger than for model 4, but again, the 95% CI's overlap by a big margin, so I can not conclude that they are different.

In model 3, which incorporates the correlation of random effects from the mother's union dissolution model, the effect on the coefficients related to residential moves is small, although they decrease a bit in size. The effect on the union dissolution variables is substantial, but still a bit smaller than when incorporating correlation from the random effects from the mother's residential moves equation or the full model 4.

It is interesting that the correlation of residential moves in model 2 has such an impact on the coefficients related to union dissolution, whereas the correlation from union dissolutions in model 3 does not. But it is not surprising, as described earlier because of the data structure, the union dissolutions are contained in the residential moves, and then the residential moves equation contains extra information on the mother's propensity to be a frequent mover.

Also, the substantial negative correlation (($\rho_{u(E,M)}$=-0.24)) between the random effects on education and residential moves was observed, while it was significantly smaller for the union dissolutions random effects ($\rho_{u(E,U)}$=-0.08).

Furthermore, the same modelling setup was used, just measuring the effect of one set of disruption indicators at a time (either residential moves or union dissolutions). This analysis showed that when leaving out one set of indicators, the reamining would show a higher impact on education. Also, the correlation of random effects from the left-out disruption equation was higher. This shows how interrelated these coefficients are, and how if not accounted for, they lead to misleading estimates.

In conclusion, not controlling for endogeneity leads to an overestimation of the effect of disruptions. For residential moves, the effect almost disappears, as we only have evidence that 3, or 5+ moves impacts the children's probability to finish secondary school more than experiencing no moves. For union dissolutions, the evidence remains that there is a negative impact from experiencing union dissolutions, but the effect decreases. Failing to account for the selection bias leads to misleading conclusions. For both disruptions related to residential moves and union dissolutions however, allowing correlations with the random effects from the equation related to mother's residential moves had the biggest impact.

The naive effect sizes from model 1 and the effect sizes after controlling for unobserved mother characteristics are similar to previous literature. Most relevant is Steele et al. (2009), who uses a similar multilevel multiprocess setup and estimates the effect of separation from the father and stepfather in a probit model setup for different educational transitions in Norway. The found magnitude of the effects before and after controlling for endogeneity are similar.

5.4 Estimated Coefficients for Event History Models

Table 7 shows the estimated coefficients for the event history model of the mother's residential moves. Overall, the results align with results from previous literature. For example, mothers with elementary education have the highest risk of moving, and mothers with a medium-level education of secondary school or diploma/bachelor have the lowest risk (Kull et al., 2016). Furthermore, the higher the move count, the more likely that the mother will move, i.e. the probability of moving increases with the number of moves that have already been done previously.

Table 8 shows the estimated coefficients for the event history model of the mother's union dissolutions. Again, the results align with previous results from the literature. The

	Model 4 $\sigma_{u(E,M)} \neq 0$, $\sigma_{u(E,U)} \neq 0, \sigma_{u(M,U)} \neq 0$	
	Mean	**95% CI**
Intercept	**-2.42**	(-2.455; -2.376)
Year	**-0.01**	(-0.037; 0.020)
Year sq	**0.04**	(0.024; 0.051)
Move count	**0.02**	(0.011; 0.025)
Move count sq	0.00	(-0.000; 0.000)
Duration (years)	**-0.62**	(-0.641; -0.591)
Duration (years) sq	**0.04**	(0.027; 0.055)
Age at move-in	**-0.49**	(-0.512; -0.471)
Age at move-in sq	**0.17**	(0.161; 0.181)
Copenhagen	**0.09**	(0.053; 0.128)
Mother's education		
Primary (Reference)	-	-
Secondary	**-0.19**	(-0.220; -0.151)
Diploma/Bachelor	**-0.16**	(-0.191; -0.120)
Postgraduate	**-0.16**	(-0.212; -0.101)

Table 7: Estimated coefficients from the random effects event-history model of residential moves, model 1 and model 4. Parameters where zero is not included in the 95% CI are in bold.

probability of the union dissolves decreases sharply if the couple is married. A higher level of education for both the mother and the partner lowers the likelihood of union dissolution, and this effect is particularly strong when the partner has postgraduate education. This also aligns with previous results, such as Steele et al. (2009) who found a similar pattern, although they estimated coefficients for different partner matchings of educational level. If the couple have mutual children, then the probability of dissolving union lowers. However, if they have children with previous partners, the probability increases, especially if the mother has children with naother partner. This was also found by Steele et al. (2009).

5.5 Rest of Covariates for Education Model

The estimated effects of the rest of the variables for the logit model for the child's educational attainment while allowing free correlation of mother-level random effects (model 4) are given in table 9. The effects are in the expected direction, and consistent with previous literature. For example boys have a lower chance of finishing secondary school, and younger cohorts have a higher probability of finishing upper secondary school. These findings are consistent with what is expected in Denmark (Birkelund, 2020; Jæger & Holm, 2007). Mother's birth age also has a substantive impact, and the older the mother, the higher probability her child has of finishing secondary school. Parents' education is also an important indicator of the child's

	Model 4 $\sigma_{u(E,M)} \neq 0$, $\sigma_{u(E,U)} \neq 0, \sigma_{u(M,U)} \neq 0$	
	Mean	**95% CI**
Intercept	**-1.53**	(-1.701; -1.369)
Year	**0.21**	(0.135; 0.295)
Year sq	**-0.29**	(-0.331; -0.253)
Union count	**-0.52**	(-0.609; -0.430)
Union count sq	**0.02**	(0.008; 0.035)
Duration (years)	**-0.44**	(-0.523; -0.336)
Duration (years) sq	**0.14**	(0.095; 0.187)
Age at union start	**-0.51**	(-0.582; -0.454)
Age at union start sq	**0.09**	(0.071; 0.110)
Married	**-0.67**	(-0.718; -0.618)
Mother's education		
Primary (Reference)	-	-
Secondary	**-0.28**	(-0.336; -0.213)
Diploma/Bachelor	**-0.52**	(-0.623; -0.416)
Postgraduate	**-0.32**	(-0.525; -0.114)
Partner's education		
Primary (Reference)	-	-
Secondary	**-0.28**	(-0.334; -0.233)
Diploma/Bachelor	**-0.33**	(-0.416; -0.230)
Postgraduate	**-0.63**	(-0.772; -0.478)
Children		
No children (Reference)	-	-
With partner	**-0.27**	(-0.301; -0.250)
With previous partner (Partner)	**0.10**	(0.038; 0.165)
With previous partner (Mother)	**0.19**	(0.103; 0.267)

Table 8: Estimated coefficients from the random effects event-history model of union dissolutions, model 1 and model 4. Parameters where zero is not included in the 95% CI are in bold.

probability to finish secondary school - the higher educated parents, the higher probability for the child. However, notice that the effect of the father's education is very similar to the mother's education. This is atypical in a lot of countries, but in Denmark, the pattern of the mother having a higher impact on the child than the father is less pronounced, perhaps because of the relatively high social mobility and shared parenting repsonsibilities between the genders (Birkelund, 2020; Jæger & Holm, 2007).

5.6 Risks to Identification Strategy

In any study involving complex social behaviors and outcomes, there are inherent risks to the validity of the identification strategy. This section outlines the potential sources of bias and limitations in the analysis, particularly concerning unmeasured characteristics, insufficient

	Model 4 $\sigma_{u(E,M)} \neq 0$, $\sigma_{u(E,U)} \neq 0, \sigma_{u(M,U)} \neq 0$	
	Mean	**95% CI**
Intercept	**-0.91**	(-1.099; -0.721)
Boy	**-1.10**	(-1.183; -1.016)
Birthyear	**0.15**	(0.102; 0.192)
Parents never married	**-0.21**	(-0.310; -0.107)
Mother's birth age		
≤21 years (Reference)	-	-
22-27 years	**0.35**	(0.206; 0.491)
28-33 years	**0.51**	(0.356; 0.655)
34+ years	**0.77**	(0.595; 0.934)
Mother's education		
Primary (Reference)	-	-
Secondary	**0.78**	(0.678; 0.887)
Diploma/Bachelor	**1.39**	(1.275; 1.501)
Postgraduate	**2.27**	(2.046; 2.488)
Father's education		
Primary (Reference)	-	-
Secondary	**0.61**	(0.500; 0.715)
Diploma/Bachelor	**1.48**	(1.349; 1.598)
Postgraduate	**1.93**	(1.739; 2.137)

Table 9: Estimated effects of other explanatory variables from the logit equation for the children's educational attainment, allowing for free correlation of mother-level random effects (Model 4). Parameters where zero is not included in the 95% CI are in bold.

within-mother variation, and data constraints.

Bias Due to Unmeasured Characteristics

A key assumption underlying this analysis is that the estimated effects of residential moves and union dissolutions on children's probability of completing high school may be biased due to selection on unmeasured characteristics attributed to mothers. These characteristics are assumed to be fixed across her different children. However, several potential sources of bias could undermine this assumption.

Firstly, the model does not account for time-varying unobservables that could influence both the likelihood of residential moves or union dissolutions and the educational outcomes of the children. For instance, if a mother gets a new, better-paying job, this may initiate a move to a different area, reducing the likelihood of subsequent moves and potentially also improving the youngest child's educational environment. This is particularly concerning if there is a significant age gap between siblings.

Secondly, the presence of child-specific unobservables. Children with the same mother may possess different inherent characteristics, such as intelligence or temperament, that affect both their educational outcomes, and their responses to familial disruptions. For example, a child with a more resilient temperament might cope better with frequent moves, resulting in less impact on their education compared to a sibling who is more sensitive to change. Additionally, parents often treat their children differently based on birth order or other factors, which could introduce variability in the children's experiences and outcomes that is not captured by the model.

Thirdly, the analysis focuses on the disruptions from the mothers side, but many children with divorced parents also spend significant time with their fathers. In Denmark, the "7-7 agreement" is common in dissoluted families, where children alternate between living with each parent every seven days. The stabilising or disrupting influences from the father's situation are not considered, which could be a source of bias. For instance, the father's involvement might mitigate some of the negative effects of disruptions. Moreover, the presence of stepparents introduces another layer of complexity, as they can significantly alter the family dynamics. This study found similar impacts of both mothers' and fathers' education on children's educational outcomes, and this suggests that the father's role is crucial and should be accounted for in a more nuanced way.

Lastly, the covariates included in the model may not be truly exogenous, potentially leading to biased estimates. If these covariates are influenced by unobserved factors that also affect the outcomes, the estimated effects of residential moves and union dissolutions could be confounded.

Insufficient Within-Mother Variation

The identification strategy relies on sufficient within-mother variation to accurately estimate mother-level random effects. While siblings share the same mother, they often experience different levels of exposure to residential moves and union dissolutions. However, the average number of children per mother in this dataset is only 1.7, and some mothers have only one child. This limited within-family variation could weaken the reliability of the mother-level random effect estimates, particularly in cases where there is insufficient exposure variability between siblings. Additionally, a significant proportion of women in the sample have just one union, which could partially explain the lower correlation observed between the education and union dissolution equations. The lack of sufficient variation in union experiences among mothers reduces the ability to accurately estimate the effects of

union dissolutions on educational outcomes. However, Steele et al. (2013) uses a similar setup with a similar multilevel structure, and although I consider it a weakness of the study, it is not a major concern.

Data Limitations

Finally, the analysis is constrained by the limitations of the available data. My model setup counters the absence of more detailed variables, such as parents' attitudes towards education, levels of familial conflict, or specific reasons for residential moves and union dissolutions. However, including more specific variables would still refine the analysis and provide a better context for understanding why these disruptions affect children's educational outcomes.

6 Conclusion

This dissertation has examined the effects of family disruptions, specifically residential moves and union dissolutions, on children's educational attainment using Danish register data and a multiprocess multilevel model with Bayesian estimation. The findings reveal that when using a simple modelling setup of multilevel modelling, both residential moves and union dissolutions have significant negative impacts on the likelihood of completing secondary education, with residential moves showing a more substantial effect. However, once endogeneity is accounted for, these effects are reduced, particularly for residential moves, where the evidence almost disappears. This highlights the importance of controlling for unobserved heterogeneity at the mother level.

The study's methodology, proves effective in addressing the selection bias in the effect of family disruptions on educational attainment. This approach provides a more nuanced understanding of how these disruptions impact educational outcomes, emphasising that simple correlations might overestimate their negative effects if endogeneity is ignored.

Future research could extend this work by incorporating a broader range of variables, such as the role of fathers or more detailed socio-demographic factors, to further refine the analysis. Additionally, following previous literature, I found weak evidence of heterogeneity effects on children's age when the disruption happens, and the gender of the child. These effects can be further investigated, especially if using a bigger sample than this study, which was limited by computational resources. Using a bigger sample would ensure sufficient data to measure the effects to get a stable estimtion.

1 Appendix A: Trace plots from MCMC samples

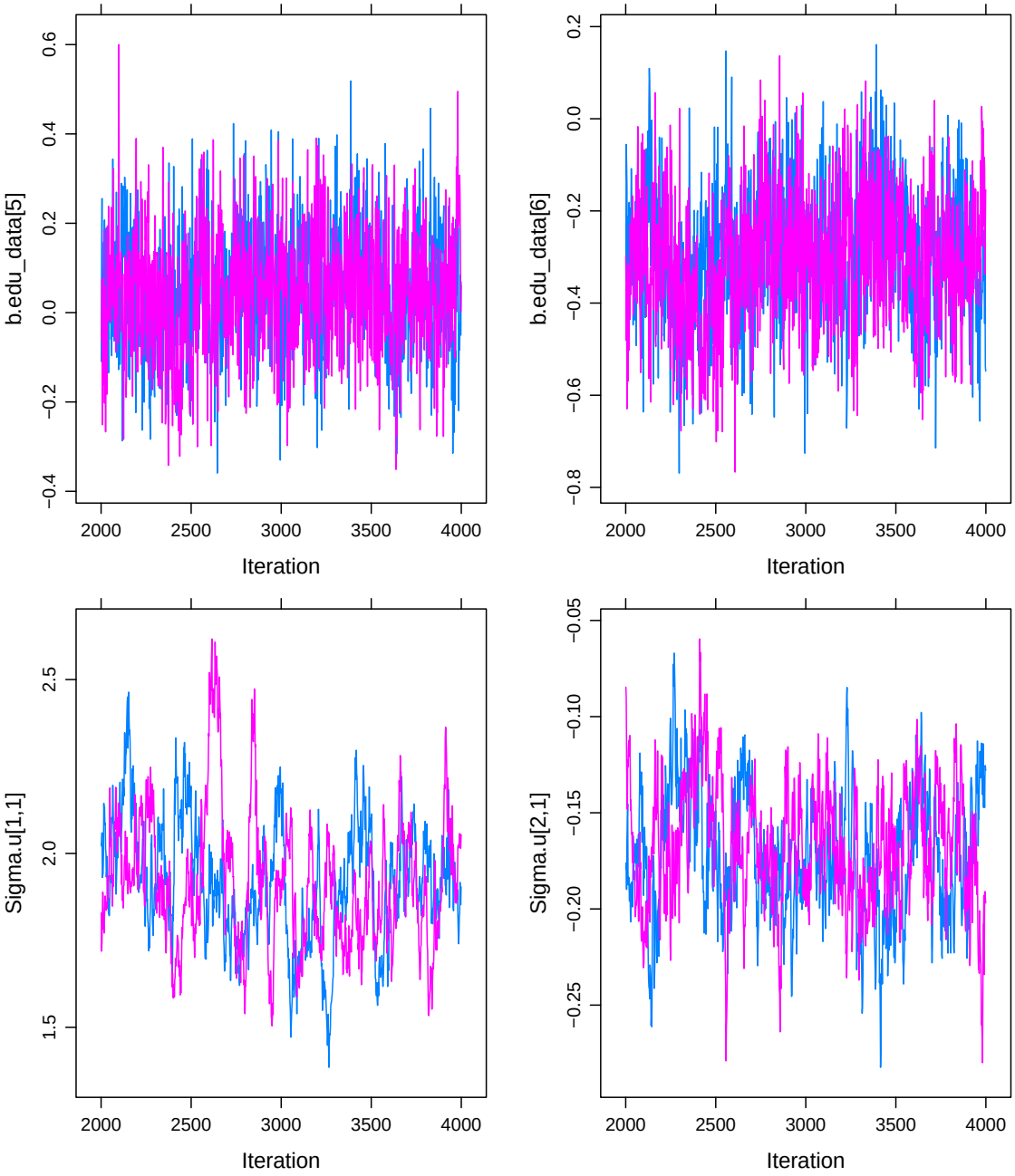


Figure 4 Trace plots from MCMC estimation of model 4. Top: Estimated effect of 1 and 2 union dissolutions on children's education. Bottom: Estimated variance of random effects for education equation, estimated covariance of random effects between education and moves equation.

2 Appendix B: Code snippet of JAGS estimation

```
# Inits ---------------------------------------
nchain=2
nsample=4000
nburnin=2000
set.seed(1234)

E_MU_R.parnames <- c(names(jags.data)[6:length(jags.data)-1],
                     rep('Sigma',3*3),rep('rho',3))

E_MU_R.par <- c('b.edu_data','b.move_data','b.union_data',
                'Sigma.u',
                'rho.u12','rho.u13','rho.u23')

E_MU_R.mod.inits <- function(chain,seed){
  b.edu_data <- c(rnorm(22,mean=0,sd=0.1))
  b.move_data <- c(rnorm(13,mean=0,sd=0.1))
  b.union_data <- c(rnorm(19,mean=0,sd=0.1))
  tau.u <- rgamma(3,shape=5)
  list("b.edu_data"=b.edu_data,
       "b.move_data"=b.move_data,
       "b.union_data"=b.union_data,
       "Tau.u"=structure(.Data = c(tau.u[1],0,0,0,tau.u[2],0,0,0,tau.u[3]), .Dim = c
    (3, 3)),
       .RNG.name = rng_name,
       .RNG.seed = seed
}

# Function to generate different initial values for each chain
rng.set <- parallel.seeds("base::BaseRNG", nchain)
initval1 <- c(E_MU_R.mod.inits(),rng.set[[1]], rng_name="lecuyer::RngStream")
initval2 <- c(E_MU_R.mod.inits(),rng.set[[2]], rng_name="base::MersenneTwister")
initval <- list(initval1,initval2)

# Run model ------------------
E_MU_R.binary_biv.jags <- run.jags('E_MU_R_binary.jag', data = jags.data,
                                   monitor=E_MU_R.par, inits=initval,
                                   n.chains = nchain, sample = nsample, thin = 1,
                                   burnin = nburnin,
                                   method='parallel',
                                   modules=c('glm','lecuyer'))
```

Listing 2: Code snippet: How the JAGS model is called in R.

```
# This defines the JAGS model in the file "E_MU_R_binary.jag"
# Logit model for three binary outcomes with correlated random effects
#----Model definition----------------
model
```

```
{
  for (i in 1:N) {
    y[i] ~ dbern(p.y[i])
    logit(p.y[i]) <-
    b.edu_data[1]*edu_data[i] +
      b.edu_data[2]*edu_data[i]*edu_move_group_1[i] +
      #... Rest of variables for education model
      b.edu_data[22]*edu_data[i]*edu_father_educat_postgraduate[i] +
      edu_data[i]*u[mother_id[i],1] +
    b.move_data[1]*move_data[i] +
      b.move_data[2]*move_data[i]*move_year[i] +
      #... Rest of variables for residential moves model
      b.move_data[12]*move_data[i]*move_mother_educat_postgraduate[i] +
      move_data[i]*u[mother_id[i],2] +
    b.union_data[1]*union_data[i] +
      b.union_data[2]*union_data[i]*union_year[i] +
      #... Rest of variables for union dissolution model
      b.union_data[18]*union_data[i]*mother_prev_children[i] +
      union_data[i]*u[mother_id[i],3]
  }
 # Specify bivariate normal random effects distribution
# Tau.u is precision matrix (inverse Sigma.u defined later)
  for (j in 1:nind) {
    u[j,1:3] ~ dmnorm(zero[1:3], Tau.u[1:3,1:3])
  }

  zero[1] <- 0
  zero[2] <- 0
  zero[3] <- 0

  # Priors for coefficients
  for (k in 1:22) {
    b.edu_data[k] ~ dnorm(0, .001)
  }
  for (k in 1:12) {
    b.move_data[k] ~ dnorm(0, .001)
  }
  for (k in 1:18) {
    b.union_data[k] ~ dnorm(0, .001)
  }

  Sigma.u <- inverse(Tau.u)
  Tau.u ~ dwish(R[1:3,1:3],3)
  rho.u12 <- Sigma.u[1,2]/sqrt(Sigma.u[1,1]*Sigma.u[2,2])
  rho.u13 <- Sigma.u[1,3]/sqrt(Sigma.u[1,1]*Sigma.u[3,3])
  rho.u23 <- Sigma.u[2,3]/sqrt(Sigma.u[2,2]*Sigma.u[3,3])
}
```

Listing 3: Code snippet: How the model is defined in JAGS.

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